Ex No:04
03.08.2026

4. SETTING UP A LOCAL AREA NETWORK.

Aim:

To design, set up, and configure a local area network connecting 4 computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command and demonstrate data sharing across the network.

IP configuration table :

Device	IP Address	subnet Mask	Connected to
PC0	125.1.1.0	255.0.0.0	Switch Fa0/1
PC1	125.1.1.1	255.0.0.0	switch Fa 0/2
PC2	125.1.1.2	255.0.0.0	switch Fa 0/3
PC3	125.1.1.3	255.0.0.0	switch Fa 0/4
switch	125.1.1.4	255.0.0.0	~~switch~~ VLAN 1 Interface

Implementation :

step 1 : open cisco packet tracer and place switch, 4 PCs and one server in the workspace.

step 2 : connect all PC's and the server to the switch using copper-straight throughth cables.

step 3 : Configure all IP addresses of all PCs and the server so that they belong to the same network.

step 4 Verify that all cable connections are active.

step 5 : Test network connectivity by opening the command prompt on a PC and using the ping command to communicate with another PC or the server.

step 6 : Enable the HTTP service on the server by selecting server → services → HTTP → ON.

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step 7: Open the web browser on a PC and enter the server's IP address to access the server's default web page

step 8: Switch from realtime mode to simulation mode in packet tracer.

step 9: Send a ping or web request from a PC to the server and use the capture/Forward or Auto Capture/Play option to observe the packet traveling through the switch.

step 10: Verify that successful communication takes place b/w the PC's and the server, demonstrating proper LAN operation and data sharing.

A Local Area Network (LAN) was successfully established using PCs, switch, and a server. Communication b/w devices was verified through ping and web access, and packet flow was observed successfully in simulation mode.